Used Car Price Estimation Using Statistical Analysis and Machine Learning in R

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***Abstract*—Pricing a used car is difficult because the selling price depends on the age, usage, engine characteristics, brand, and condition of the vehicle. This paper presents a data-driven price-estimation study implemented in the R programming environment using the CarDekho used-car dataset. After removing 167 duplicate records, 15,244 listings with 13 original fields were retained. Exploratory analysis, interquartile-range outlier screening, a correlation study, and a Kruskal–Wallis test were used to understand the data. Two models were then trained on the logarithm of the selling price using an 80/20 train–test split: multiple linear regression and a regression tree. On the held-out test set the regression tree obtained an RMSE of Rs. 406,712, an MAE of Rs. 149,761, and an R^2 of 0.760, compared with Rs. 431,363, Rs. 151,117, and 0.730 for linear regression. Maximum power, engine capacity, brand, and vehicle age were the most influential variables. Errors were largest for expensive and rare vehicles, and the regression tree cannot predict above the highest price level seen in its leaves. The study evaluates price-prediction accuracy on one platform's listings and is not a valuation tool for individual vehicles.**

***Index Terms*—Used car price prediction, regression tree, linear regression, exploratory data analysis, log transformation, Kruskal–Wallis test, R programming.**

I. Introduction

The second-hand car market is large and fragmented. Buyers and sellers usually do not know the fair price of a particular vehicle, and an incorrect asking price can either delay a sale or cause the seller to lose money. The price of a used car is influenced by measurable attributes such as age, distance driven, engine size, power output, fuel type, and transmission, as well as by brand reputation. These attributes can be recorded in a structured table, which makes the problem suitable for statistical modelling.

Machine learning methods can learn the relationship between such attributes and the observed selling prices from past listings and then estimate the price of listings that were not used for training. Different algorithms make different assumptions. Linear regression assumes an additive relationship between predictors and the response, whereas a regression tree divides the data into groups using a sequence of threshold rules and can represent nonlinear effects and interactions.

This study analyses a used-car listing dataset in R. The work follows the full workflow from data loading and cleaning to exploratory analysis, hypothesis testing, model training, and evaluation. The main objectives are as follows. The first is to describe the structure and quality of the dataset. The second is to identify which attributes are associated with the selling price. The third is to compare a linear regression model and a regression tree on unseen data. The fourth is to interpret the results and identify the limitations of the approach.

The contribution is an empirical and reproducible analysis of a small, interpretable price-estimation pipeline. The study does not claim to build a production valuation system, and the results apply to the data and settings reported.

II. Problem Formulation and Objectives

Let a listing i be described by a feature vector

$$\mathbf{x}_i = [x_{i1}, x_{i2}, \ldots, x_{i,10}], \tag{1}$$

where the ten features are vehicle age, kilometres driven, seller type, fuel type, transmission type, mileage, engine capacity, maximum power, number of seats, and brand. Let p_i be the selling price in rupees. The dataset is

$$D = \{(\mathbf{x}_i, p_i)\}_{i=1}^{N}, \tag{2}$$

with $N = 15{,}244$ after cleaning. Because prices are strongly right-skewed, the model target is

$$y_i = \ln(p_i), \tag{3}$$

and a predicted log price $\hat{y}_i$ is converted back to rupees using

$$\hat{p}_i = \max\bigl(0, \exp(\hat{y}_i)\bigr). \tag{4}$$

The learning task is to find a function f such that $\hat{y}_i = f(\mathbf{x}_i)$ generalises to listings that were not used to fit the model.

The specific objectives are to examine the data, test whether prices differ across fuel types, compare two regression models using rupee-scale test metrics, and interpret the fitted models. The scope is limited to price estimation from the listed attributes. The study does not include inspection of vehicle condition, accident history, location, or market trends over time.

III. Related Work

Price prediction for second-hand cars has been studied with several machine-learning techniques. Pudaruth compared multiple linear regression, k-nearest neighbours, decision trees, and naive Bayes for predicting used-car prices from advertised

TABLE I
DATASET SUMMARY

Parameter	Reported Value
Rows read	15,411
Exact duplicates removed	167
Invalid rows removed	0
Rows modelled	15,244
Original fields	13
Model input features	10
Brands	32
Training rows (80%)	12,195
Testing rows (20%)	3,049
Missing values after loading	0

attributes [1]. Gegic et al. applied a combination of machine-learning methods to car-price prediction and reported on the practical value of such models for buyers and sellers [2]. These studies indicate that conventional supervised learning on tabular attributes is a reasonable baseline for this problem.

Classification and Regression Trees (CART), introduced by Breiman et al., split the feature space recursively into regions with similar responses, and the resulting model is easy to interpret [3]. Ensemble extensions such as random forests combine many trees to reduce variance [4]. Hastie et al. discuss the trade-off between the interpretability of linear models and trees and the flexibility of more complex learners [5]. The regression tree in this work is fitted with the `rpart` package [7], which implements the CART approach in R [8].

The Kruskal–Wallis test is a rank-based method for comparing a continuous variable across more than two groups without assuming normality [6]. It is used here because the price distribution is far from normal.

The present study differs from work using larger ensembles or deep models because it deliberately uses two simple, inspectable models. It is therefore a baseline analysis and not an attempt to reproduce the best accuracy obtainable on this data.

IV. DATASET DESCRIPTION

The dataset is the CarDekho used-car listing dataset [9]. The file contains 15,411 rows. After removing the exported row-index column, the original fields are car name, brand, model, vehicle age, kilometres driven, seller type, fuel type, transmission type, mileage, engine capacity, maximum power, number of seats, and selling price. Table I gives a summary of the cleaned data.

No missing values were present in the cleaned data. Of the listings, 9,459 were sold by dealers, 5,612 by individuals, and 173 by Trustmark dealers. Manual transmission accounts for 12,094 listings and automatic for 3,150. The two main fuel types are petrol (7,555) and diesel (7,342), with small numbers of CNG (299), LPG (44), and electric (4) vehicles. The most frequent brands are Maruti (4,933), Hyundai (2,952), Honda (1,476), Mahindra (999), Toyota (789), and Ford (776).

Table II summarises the numeric fields. Several values deserve caution. The maximum distance driven is 3.8 million km, which is implausible for a passenger car and is probably an entry error, and some listings report zero seats. The script does not remove such records because they were not clearly invalid under the rules defined in advance (non-positive price, negative age, negative distance). The selling price ranges from Rs. 40,000 to Rs. 39.5 million, so the data include a small number of very expensive vehicles.

TABLE II
SUMMARY OF NUMERIC FIELDS AFTER CLEANING

Feature	Min	Median	Mean	Max	IQR outl.
Vehicle age (yr)	0	6	6.04	29	154
Km driven	100	50,000	55,640	3.8M	464
Mileage (km/l)	4.0	19.67	19.70	33.54	87
Engine (cc)	793	1,248	1,486	6,592	2,103
Max power (bhp)	38.4	88.5	100.6	626	853
Seats	0	5	5.33	9	2,477
Price (Rs.)	40,000	559,000	774,701	39.5M	1,371

V. PROPOSED METHODOLOGY

The workflow consists of data loading, cleaning, outlier screening, exploratory analysis, statistical testing, feature preparation, a train–test split, model training, and evaluation. All steps were implemented in a single R script with a fixed random seed (42) so that the results can be reproduced.

A. Data Cleaning

The script checks that all 13 required fields are present and converts the numeric fields to numbers. Exact duplicate rows are removed, which eliminated 167 records. Records with a non-positive price, a negative age, or a negative distance would also be removed, but none were found. Missing numeric values would be replaced by the median and missing categorical values by the most frequent category, although the cleaned data contained no missing entries.

B. Outlier Screening

Outliers are identified with the interquartile-range (IQR) rule. A value x is flagged if

$$x < Q_1 - 1.5\,\mathrm{IQR} \quad \text{or} \quad x > Q_3 + 1.5\,\mathrm{IQR}, \tag{5}$$

where $\mathrm{IQR} = Q_3 - Q_1$. The outliers are counted and reported, but they are not removed, because high-priced and high-power cars are genuine listings that a price model may need to handle. The counts are given in Table II. Notably, 1,371 listings (about 9.0%) are price outliers. The large count for seats arises because almost all vehicles have five seats, so any other value is flagged by the rule.

C. Feature Preparation

The ten model inputs are vehicle age, kilometres driven, seller type, fuel type, transmission type, mileage, engine capacity, maximum power, number of seats, and brand. The car name and model fields are excluded because they contain many unique categories that would make the models unstable. Seller type, fuel type, transmission type, and brand are converted to categorical factors. The response is the natural logarithm of the selling price to reduce the effect of skewness.

D. Train–Test Split

The cleaned data are randomly divided into 80% for training (12,195 listings) and 20% for testing (3,049 listings). The models are fitted only on the training data, and all reported test metrics use the held-out data. A random split does not guarantee that near-identical listings (for example, the same model and year sold by different sellers) fall on the same side of the split, so the reported performance may be somewhat optimistic.

E. Linear Regression

The first model is a multiple linear regression of log price on all ten features. Its coefficients are interpretable: because the response is a logarithm, a coefficient β corresponds to an approximate multiplicative change of e^{β} in price for a one-unit increase in the predictor.

F. Regression Tree

The second model is a regression tree fitted with the `rpart` package using the ANOVA method. The complexity parameter is set to 0.001, the minimum number of observations required to attempt a split is 30, and the maximum depth is 8. The fitted tree has 73 nodes and 37 leaves. The tree predicts the average log price of the training listings in the leaf that a vehicle reaches, so its predictions take only a limited number of distinct values. No pruning by cross-validation or hyperparameter search was performed.

VI. Evaluation Metrics

Predictions are converted back to rupees, and performance is measured on the test set with three metrics. The root mean squared error is

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(p_i - \hat{p}_i)^2}, \tag{6}$$

the mean absolute error is

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|p_i - \hat{p}_i|, \tag{7}$$

and the coefficient of determination is

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(p_i - \hat{p}_i)^2}{\sum_{i=1}^{n}(p_i - \bar{p})^2}. \tag{8}$$

RMSE penalises large errors heavily and is therefore sensitive to expensive vehicles, whereas MAE gives equal weight to each rupee of error. The model with the lower RMSE is selected as the final model by the script. As supplementary checks on the same test split, the root mean squared error on the log scale, the log-scale R^2, the mean absolute percentage error (MAPE), and the median absolute percentage error (MdAPE) were also computed, because a few very expensive cars can dominate the rupee-scale metrics.

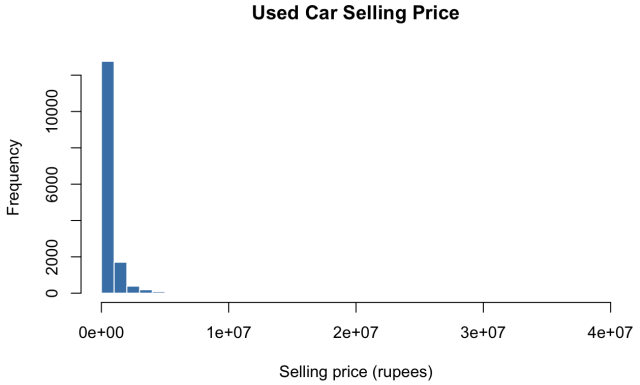


Fig. 1. Distribution of the used-car selling price.

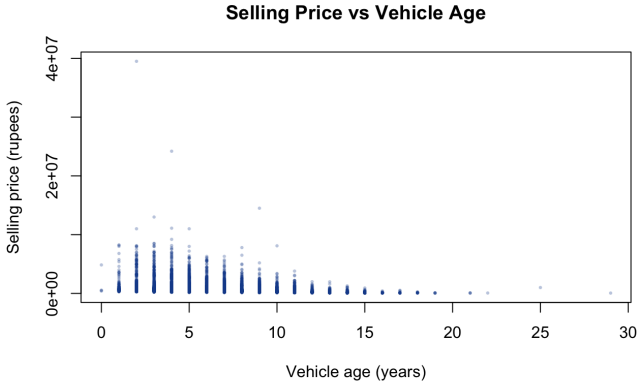


Fig. 2. Selling price versus vehicle age.

VII. Results and Discussion

A. Exploratory Analysis

Fig. 1 shows that the selling price is strongly right-skewed. Most vehicles are sold for a few hundred thousand rupees, and a long tail extends to about Rs. 40 million. This motivates the log transformation of the response. Fig. 2 shows that prices tend to fall as the vehicle becomes older. Listings aged between about two and five years show the widest range of prices, and prices beyond 15 years are uniformly low. A few high-priced points at ages of 2, 4, and 9 years correspond to luxury vehicles.

Fig. 3 and Table III compare prices across fuel types. Diesel vehicles have the highest median price among the two large groups (Rs. 700,000 against Rs. 460,000 for petrol), and both groups contain many high-price outliers. The electric group contains only four vehicles, so its mean should not be regarded as representative. LPG vehicles are the cheapest on average.

The correlation matrix in Fig. 4 shows that the selling price is most strongly and positively correlated with maximum power ($r = 0.751$) and engine capacity ($r = 0.586$). It is negatively correlated with mileage ($r = -0.305$) and vehicle age ($r = -0.241$), and only weakly related to kilometres

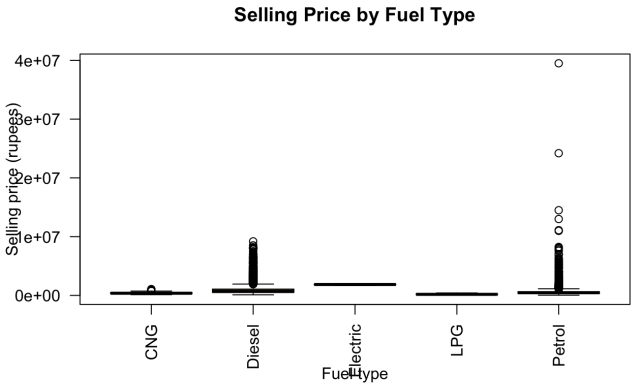


Fig. 3. Selling price by fuel type.

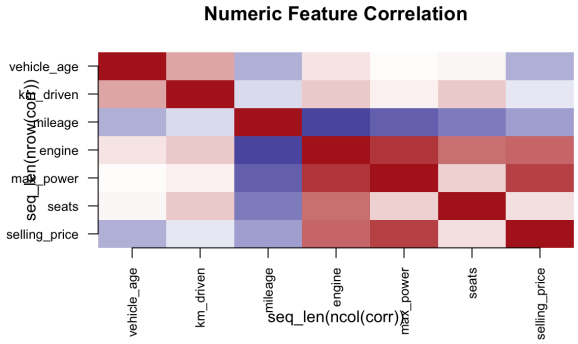


Fig. 4. Correlation matrix of the numeric features.

TABLE III
SELLING PRICE (RS.) BY FUEL TYPE

Fuel	Count	Mean	Median	Std. dev.
CNG	299	417,689	370,000	180,065
Diesel	7,342	1,000,458	700,000	947,552
Electric	4	1,853,500	1,857,500	123,122
LPG	44	206,273	182,500	83,080
Petrol	7,555	572,178	460,000	802,019

TABLE IV
TEST-SET MODEL COMPARISON (RUPEE SCALE)

Model	RMSE	MAE	R^2
Linear regression (log price)	431,363	151,117	0.7299
Regression tree (log price)	406,712	149,761	0.7599

driven ($r = -0.080$) and seats ($r = 0.115$). Engine capacity and maximum power are themselves highly correlated ($r = 0.807$), and mileage is negatively correlated with engine capacity ($r = -0.632$). This overlap among predictors means that individual linear-regression coefficients for these variables should be interpreted with care.

B. Statistical Test

A Kruskal–Wallis test was used to check whether selling prices differ across fuel types. The test statistic was $H = 2632.5$ with 4 degrees of freedom and a p-value below 2.2×10^{-16}. The null hypothesis of identical price distributions across fuel types is therefore rejected. This shows that price is associated with fuel type, but it does not show that fuel type itself causes the difference, because fuel type is related to other attributes such as engine size, power, and brand. The test does not indicate which pairs of fuel types differ, and the electric and LPG groups are very small.

C. Model Comparison

Table IV presents the test-set results. The regression tree achieved a lower RMSE (Rs. 406,712 against Rs. 431,363), a slightly lower MAE (Rs. 149,761 against Rs. 151,117), and a higher R^2 (0.760 against 0.730) than linear regression. It was therefore selected as the final model by the RMSE rule. The RMSE improvement is approximately 5.7%.

The advantage of the tree is not uniform across all measures. Table V shows that on the log scale, linear regression achieved a lower error and a higher R^2, and its median percentage error was lower. The tree therefore performs better on the expensive vehicles that dominate the rupee-scale RMSE, while linear regression is slightly better for typical vehicles. The two models have comparable MAPE values of about 18.5%, and the median percentage error of roughly 14% indicates that a typical prediction is off by about one-seventh of the actual price. The difference between the models is modest, and with a single split it cannot be considered conclusive.

Fig. 5 plots the predicted price of the tree against the actual price. The points are close to the red line of perfect prediction for vehicles below about Rs. 2 million, which is where most of the data lie. The horizontal bands are characteristic of a regression tree, which assigns the same prediction to every vehicle in a leaf. For high-priced vehicles the points fall mostly below the line, and the highest predictions flatten near Rs. 7 million, so the model underestimates luxury vehicles, including those whose actual prices reach Rs. 13–14 million.

D. Interpretation of the Models

The linear model achieved an R^2 of 0.897 on the log-price training data. Table VI lists selected coefficients. Holding other variables constant, each additional year of age is associated with an approximately 11.6% lower price ($e^{-0.1232} = 0.884$), and each additional bhp of maximum power with an approximately 0.7% higher price. Manual transmission is associated with a price about 10% lower than automatic, and sales by individuals with a price about 7% lower than dealer sales. Diesel vehicles are priced about 24% higher than CNG vehicles, which is the reference fuel type. The coefficient for electric vehicles is based on only four listings, and the LPG coefficient is not statistically significant ($p = 0.104$). The brand coefficients are not shown.

TABLE V
SUPPLEMENTARY TEST-SET METRICS

Model	Log RMSE	Log R^2	MAPE (%)	MdAPE (%)
Linear regression	0.2331	0.8842	18.54	13.84
Regression tree	0.2415	0.8757	18.66	14.40

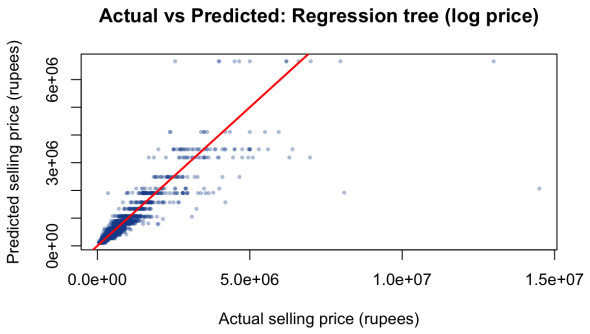


Fig. 5. Actual versus predicted selling price for the selected regression tree on the test set.

For the regression tree, the first split is on maximum power, and the next splits use vehicle age and engine capacity. The variable-importance scores reported by `rpart` are shown in Table VII. Maximum power (3,634.6), engine capacity (2,265.9), brand (1,911.8), and vehicle age (1,389.4) are the most important, followed by mileage (707.0). Kilometres driven, seats, and fuel type have comparatively little importance. These scores include surrogate splits, so correlated variables such as engine and power share credit. The ranking agrees with the correlation analysis, in which maximum power and engine capacity had the strongest relationship with price.

Importance and regression coefficients describe associations in the data. They do not establish that changing one attribute of a vehicle would cause its price to change by the stated amount.

VIII. STATISTICAL ANALYSIS

The Kruskal–Wallis statistic is computed from the ranks of the prices across the k fuel groups as

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1), \tag{9}$$

where R_j is the rank sum and n_j the size of group j [6]. The test strongly rejects equal price distributions across fuel types ($H = 2632.5$, $p < 2.2 \times 10^{-16}$).

The comparison between the two models is descriptive. The models were evaluated on one random split, and no confidence intervals, paired test on the prediction errors, or repeated cross-validation was used. The difference in R^2 of about 0.03 should therefore be read as the result for this particular split and seed rather than as a statistically established difference.

TABLE VI
SELECTED LINEAR REGRESSION COEFFICIENTS (RESPONSE: LOG PRICE)

Predictor	Estimate	Std. error	p-value
Vehicle age (per year)	−0.1232	0.0008	< 0.001
Max power (per bhp)	0.0067	0.0001	< 0.001
Engine (per cc)	0.00012	0.00001	< 0.001
Mileage (per km/l)	−0.0157	0.0010	< 0.001
Seats (per seat)	0.0401	0.0046	< 0.001
Manual transmission	−0.1079	0.0065	< 0.001
Individual seller	−0.0708	0.0044	< 0.001
Diesel	0.2178	0.0151	< 0.001
Petrol	−0.0461	0.0158	0.003
LPG	−0.0633	0.0390	0.104
Electric	0.4232	0.1291	0.001

TABLE VII
REGRESSION TREE VARIABLE IMPORTANCE (`RPART`)

Feature	Importance
Maximum power	3,634.6
Engine capacity	2,265.9
Brand	1,911.8
Vehicle age	1,389.4
Mileage	707.0
Kilometres driven	148.3
Seats	138.2
Fuel type	116.3

IX. LIMITATIONS

The experiment has several limitations. First, only one random train–test split with a single seed was used, and the regression tree was not pruned or tuned, so the ranking of the two models could change with a different split or settings. Second, the split does not account for near-duplicate listings, which may inflate the performance estimate. Third, the data include implausible values such as a distance of 3.8 million km and listings with zero seats, and these were kept in the analysis. Fourth, the price distribution has a heavy tail with few very expensive vehicles; both models, and the tree in particular, underestimate these listings, and the rupee-scale RMSE is dominated by them. Fifth, the electric and LPG groups contain very few vehicles, so conclusions about them are unreliable.

Sixth, the data come from a single platform and record a selling price without information about condition, accident history, ownership count, service history, location, or the date of sale, all of which affect real prices. The model cannot capture inflation or market changes over time. Finally, car name and model were excluded, so the model uses brand but not the specific model, which limits its ability to distinguish vehicles of the same brand. The results should therefore not be used as a replacement for professional valuation.

X. FUTURE WORK

Several extensions could improve the study. The first is to use k-fold cross-validation and report confidence intervals

for the metrics, and to prune the tree using the complexity-parameter table. The second is to evaluate ensemble methods such as random forests and gradient-boosted trees, and regularised regression, which may address the stepwise predictions and the underestimation of expensive cars. The third is to include the car model through grouping or target encoding and to add interaction terms such as age and brand. The fourth is to investigate the implausible records and the outliers, for example by comparing results with and without them or by using robust loss functions. The fifth is to use a time-aware split, or listing-level grouping, to avoid leakage between near-identical records. Finally, adding condition, location, and sale-date information would be necessary before the approach could be considered for real use.

XI. Conclusion

This paper presented a used-car price-estimation study in R using 15,244 cleaned listings from the CarDekho dataset. Exploratory analysis showed a heavily right-skewed price distribution, a decline in price with age, and the strongest positive association of price with maximum power and engine capacity. A Kruskal–Wallis test showed that price differs significantly across fuel types. A regression tree trained on log price achieved an RMSE of Rs. 406,712, an MAE of Rs. 149,761, and an R^2 of 0.760 on the test set, slightly better than linear regression on the rupee-scale metrics, although linear regression was slightly better on the log scale and in median percentage error. Maximum power, engine capacity, brand, and age were the dominant predictors.

The results support the use of simple, interpretable models as a baseline for used-car pricing, while showing that expensive vehicles are poorly predicted and that the comparison between the two models is based on a single split. Cross-validation, ensemble models, cleaner data, and additional features such as condition and location are needed before the approach could be relied upon in practice.

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